

Lecture # 4

Thursday
14 March 2024
10-11 AM
3-4 PM

PH102

After Mid Semester Exams

by

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Chapter-5

Magnetostatics

Magnetic Moment
Magnetic Vector Potential

Magnetic Moment

The product $I\vec{A}$ is defined to be the magnetic dipole moment $\vec{m}$, called ‘Magnetic Moment’ of the loop.

$$\vec{m} = I\vec{a}$$

$$\vec{\tau} = \vec{m} \times \vec{B}$$

It is well known that, the **torque** exerted on a loop carrying current I , placed in a magnetic field $\vec{B}$ is:

$$\vec{\tau} = I\vec{a} \times \vec{B}$$

Here, $\vec{a}$ is the area vector with $|a|$ is the area of the loop.

The Magnetic field due to a Magnetic dipole

The vector potential due to small element $d\mathbf{l}'$ at a point P is

$$\overrightarrow{A(r)} = \frac{\mu_0}{4\pi} \oint_C \frac{\vec{I}}{r} dl$$

The integral is transformed via the corollary of Stokes' theorem. Page-56, Problem 1.60.

$$\oint_S \overrightarrow{da} \times \nabla f = \oint_C f d\mathbf{l}$$

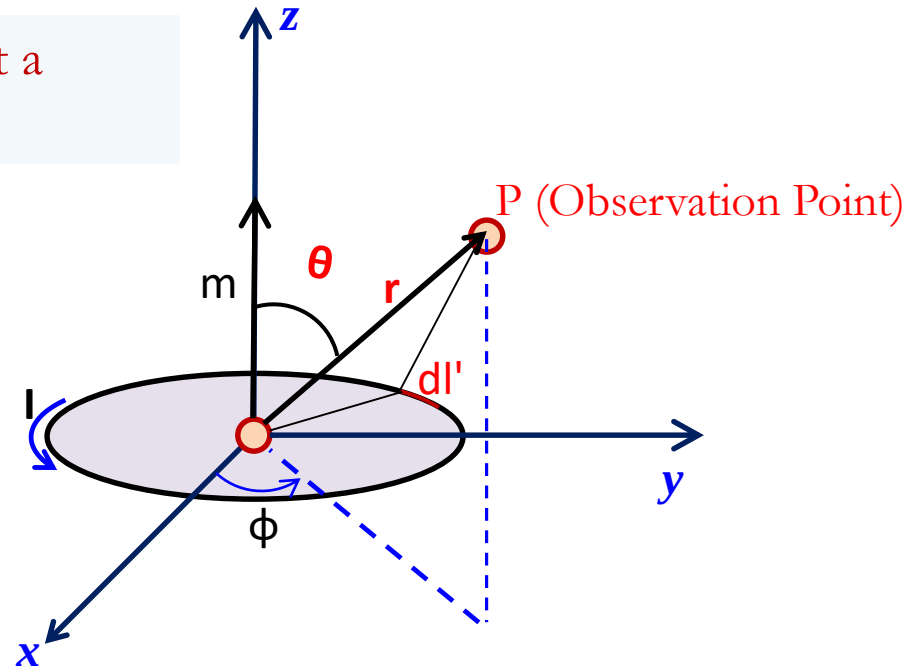
$$\overrightarrow{A(r)} = \frac{\mu_0 I}{4\pi} \oint_S \overrightarrow{da} \times \nabla \left(\frac{1}{r} \right)$$

$$\overrightarrow{A(r)} = \frac{\mu_0}{4\pi} \oint_S \overrightarrow{dm} \times \left(\frac{\hat{r}}{r^2} \right)$$

$$\nabla \left(\frac{1}{r} \right) = \frac{\hat{r}}{r^2}$$

Here

$$I d\mathbf{a}' = d\mathbf{m}$$



A "small" current loop now means that the term $\frac{\hat{r}}{r^2}$ remains essentially constant for all $d\mathbf{m}$, so it can be factored from the integral.

$$\vec{A}(r) = \frac{\mu_0}{4\pi} \left[\oint_s d\vec{m} \right] \times \left(\frac{\hat{r}}{r^2} \right)$$

$$\vec{A}(r) = \frac{\mu_0}{4\pi} \left(\vec{m} \times \left(\frac{\hat{r}}{r^2} \right) \right)$$

Here $\mathbf{m}$ is the magnetic dipole moment of the small loop

Now the Magnetic field $\mathbf{B}$ is give by the relation

$$\vec{B} = \nabla \times \vec{A}$$

$$\vec{B} = \nabla \times \frac{\mu_0}{4\pi} \left(\vec{m} \times \left(\frac{\hat{r}}{r^2} \right) \right)$$

$$\vec{B} = \nabla \times \frac{\mu_0}{4\pi} \left(\vec{m} \times \left(\frac{\hat{r}}{r^2} \right) \right)$$

$$\vec{m} = m\hat{z}$$

$$\vec{B} = \nabla \times \frac{\mu_0}{4\pi} \left(m\hat{z} \times \left(\frac{\hat{r}}{r^2} \right) \right)$$

$$\vec{B} = \frac{\mu_0 m}{4\pi} \left(\nabla \times \left(\hat{z} \times \left(\frac{\hat{r}}{r^2} \right) \right) \right)$$

$$\hat{z} \times \left(\frac{\hat{r}}{r^2} \right) = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ 0 & 0 & \omega \\ \sin \theta \cos \phi / r^2 & \sin \theta \sin \phi / r^2 & \cos \theta / r^2 \end{vmatrix}$$

$$= 1/r^2 [-(\sin \theta \sin \phi) \hat{x} + (\sin \theta \cos \phi) \hat{y}]$$

$$= \sin \theta / r^2 [-(\sin \phi) \hat{x} + (\cos \phi) \hat{y}]$$

$$\hat{z} \times \left(\frac{\hat{r}}{r^2} \right) = \frac{\sin \theta}{r^2} \hat{\phi}$$

$$\hat{\phi} = -(\sin \phi) \hat{x} + (\cos \phi) \hat{y}$$

Now the curl of this function is

$$\nabla \times \frac{\sin \theta}{r^2} \hat{\phi}$$

The curl in the spherical co-ordinates is defined as

Zero since no θ component

Zero since no r component

$$\begin{aligned} \nabla \times \vec{v} = & \frac{1}{r \sin \theta} \left[\frac{\partial}{\partial \theta} (\sin \theta v_{\phi}) - \frac{\partial v_{\theta}}{\partial \phi} \right] \hat{r} + \frac{1}{r} \left[\frac{1}{\sin \theta} \frac{\partial v_r}{\partial \phi} - \frac{\partial}{\partial r} (r v_{\phi}) \right] \hat{\theta} \\ & + \frac{1}{r} \left[\frac{\partial}{\partial r} (r v_{\theta}) - \frac{\partial v_r}{\partial \theta} \right] \hat{\phi} \end{aligned}$$

$$\frac{\partial}{\partial \theta} (\sin \theta v_\phi) = \frac{2 \sin \theta \cos \theta}{r^2}$$

$$\frac{\partial}{\partial r} (r v_\phi) = -\frac{\sin \theta}{r^2}$$

$$\nabla \times \frac{\sin \theta}{r^2} \hat{\phi} = \frac{1}{r^3} [2 \cos \theta \hat{r} + \sin \theta \hat{\theta}]$$

$$\vec{B} = \frac{\mu_0 m}{4\pi} \frac{1}{r^3} [2 \cos \theta \hat{r} + \sin \theta \hat{\theta}]$$

$$\vec{B} = \frac{\mu_0 m}{4\pi} \left(\nabla \times \left(\hat{z} \times \left(\frac{\hat{r}}{r^2} \right) \right) \right)$$

In the co-ordinate free form this equation can be written as

$$\vec{m} = m \hat{z} = m [\cos \theta \hat{r} - \sin \theta \hat{\theta}]$$

$$\vec{m} = [(m \cdot \hat{r}) \hat{r} - (m \cdot \hat{\theta}) \hat{\theta}]$$

$$\vec{m} = [(\vec{m} \cdot \hat{r})\hat{r} - (\vec{m} \cdot \hat{\theta})\hat{\theta}]$$

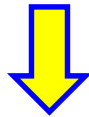
Consider the term $3(\vec{m} \cdot \hat{r})\hat{r} - \vec{m}$

$$3(\vec{m} \cdot \hat{r})\hat{r} - \vec{m} = 3m \cos \theta \hat{r} + m \sin \theta \hat{\theta} - m \cos \theta \hat{r}$$

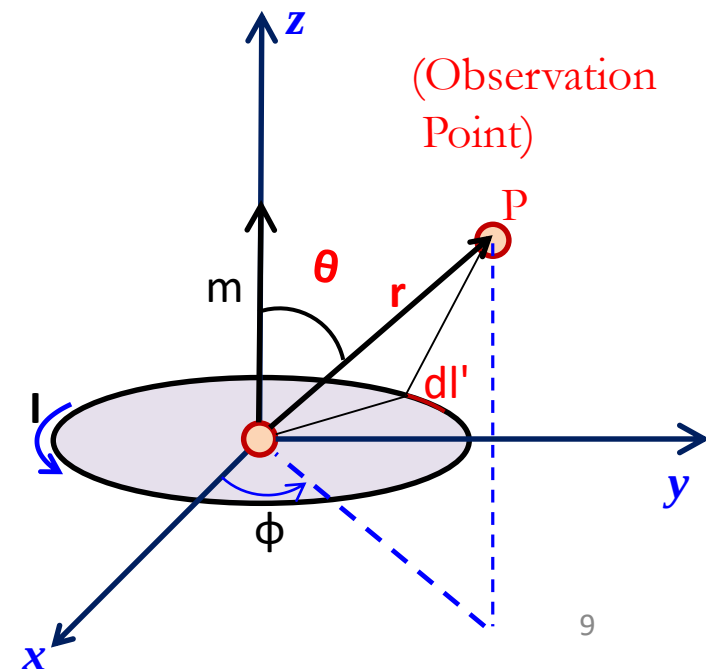
$$3(\vec{m} \cdot \hat{r})\hat{r} - \vec{m} = 2m \cos \theta \hat{r} + m \sin \theta \hat{\theta}$$



$$\vec{B} = \frac{\mu_0 m}{4\pi r^3} [2 \cos \theta \hat{r} + \sin \theta \hat{\theta}]$$



$$\vec{B} = \frac{\mu_0}{4\pi r^3} [3(\vec{m} \cdot \hat{r})\hat{r} - \vec{m}]$$



$$\vec{B} = \frac{\mu_0 m}{4\pi} \frac{1}{r^3} [2 \cos \theta \hat{r} + \sin \theta \hat{\theta}]$$

$$\vec{B} = \frac{\mu_0}{4\pi} \frac{1}{r^3} [3(\vec{m} \cdot \hat{r})\hat{r} - \vec{m}]$$

These results are analogues to the field of an Electric dipole

$$\vec{E}(r, \theta) = \frac{p}{4\pi\epsilon_0} \frac{1}{r^3} [2 \cos \theta \hat{r} + \sin \theta \hat{\theta}]$$

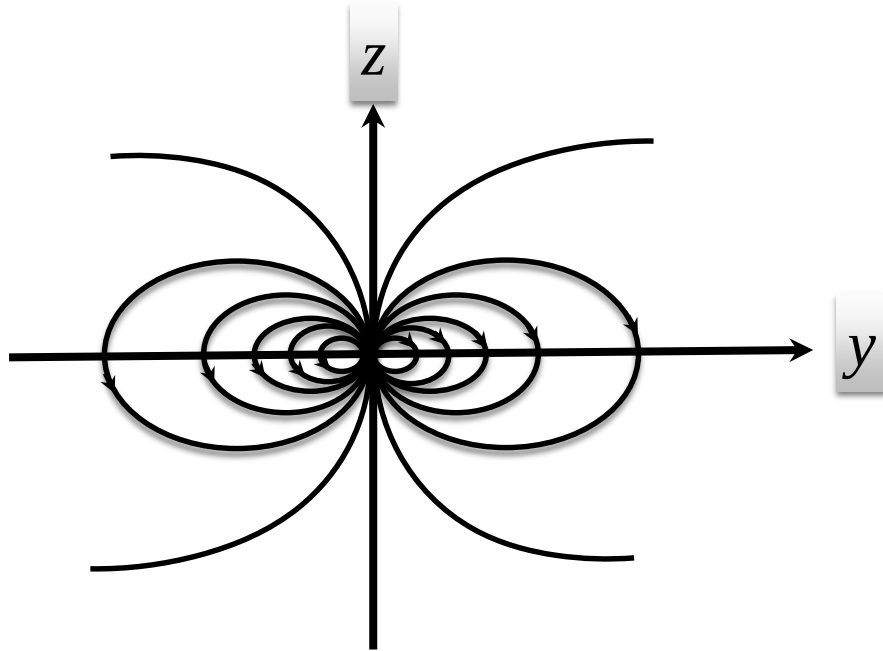
$$\vec{E}(r) = \frac{1}{4\pi\epsilon_0} \frac{1}{r^3} [3(\vec{p} \cdot \hat{r})\hat{r} - \vec{p}]$$

Summarized Results

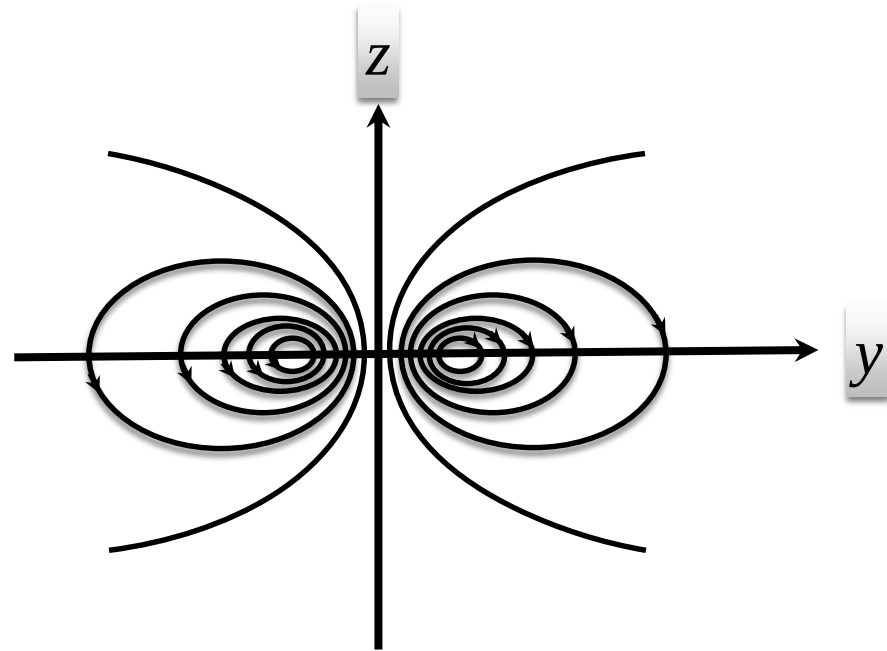
$$\overrightarrow{A(r)} = \frac{\mu_0}{4\pi} \left(\frac{m \sin \theta}{r^2} \right) \hat{\phi}$$

$$\vec{B} = \frac{\mu_0 m}{4\pi} \frac{1}{r^3} [2 \cos \theta \hat{r} + \sin \theta \hat{\theta}]$$

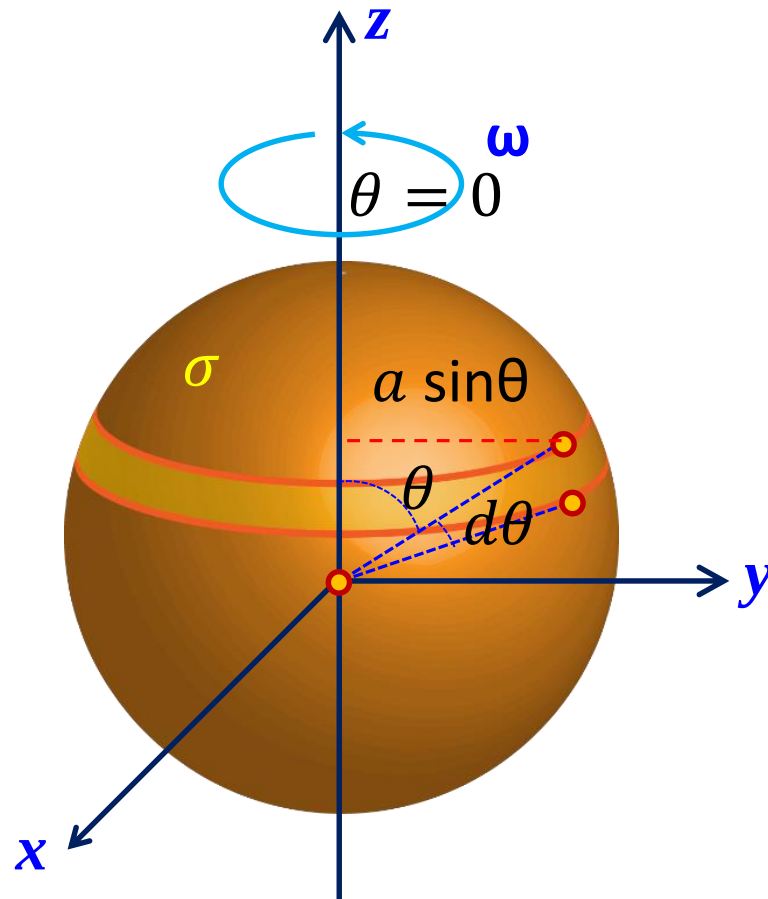
Field of a "pure" dipole



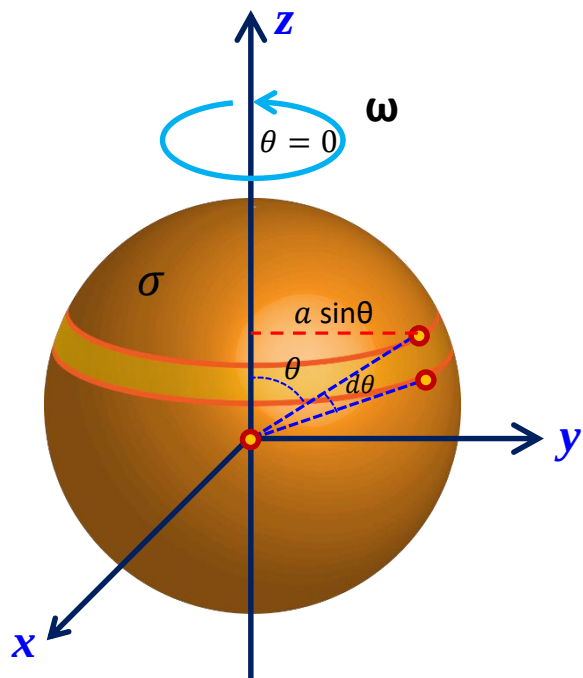
Field of a "physical" dipole



Example: A uniform surface charge density ' σ ' exists on a sphere of radius ' a '. Calculate the equivalent magnetic dipole moment of the sphere which is rotating with angular velocity ' ω ' about a diameter.



Solution: The shaded ring between θ and $\theta + d\theta$ has width $a d\theta$ and radius $a \sin \theta$.



This ring's area $= (2\pi a \sin \theta)(a d\theta)$

$$= 2\pi a^2 \sin \theta d\theta$$

Charge on it $= 2\pi \sigma a^2 \sin \theta d\theta$

Now, What is the equivalent current if the angular velocity ω is about the line $\theta = 0$

Let us consider an observer who is stationary on the laboratory and is observing the shaded ring as it rotates. The total charge on the ring is $2\pi \sigma a^2 \sin \theta d\theta$ and it rotates past the observer $\frac{\omega}{2\pi}$ times in unit time.

Therefore the current is

$$I = \frac{2\pi\sigma a^2 \sin\theta a d\theta \frac{\omega}{2\pi}}{\omega\sigma a^2 \sin\theta a d\theta}$$

$$\text{Since, } T = \frac{2\pi}{\omega} \text{ \& } I = \frac{Q}{T}$$

The magnetic dipole moment of the elementary strip (ring)

$$dm = \text{current} \times \text{Area} \quad \text{Loop Area}$$

$$dm = \omega\sigma a^2 \sin\theta a d\theta \times \pi a^2 \sin^2\theta$$

Thus the total dipole moment is:

$$m = \int dm$$

$$m = \int_0^\pi \pi\omega\sigma a^4 \sin^3\theta d\theta$$

$$m = \int_0^\pi \pi \omega \sigma a^4 \sin^3 \theta \, d\theta$$

Since

$$\int_0^\pi \sin^3 \theta \, d\theta = \frac{4}{3}$$

$$m = \frac{1}{3} (4\pi a^2 \sigma) \omega a^2$$

$$m = \frac{1}{3} Q \omega a^2$$

Where

Total Charge

$$Q = 4\pi a^2 \sigma$$

Hence the magnetic moment

$$m = \frac{1}{3} Q \omega a^2$$

Example

A uniform surface charge density σ exists on a hemispherical shell of radius R . Calculate the magnetic field induction $\vec{B}$ at point P if the hemisphere is rotating with constant angular velocity ω about the axis passing through 'P' and the centre 'C' as shown in figure.

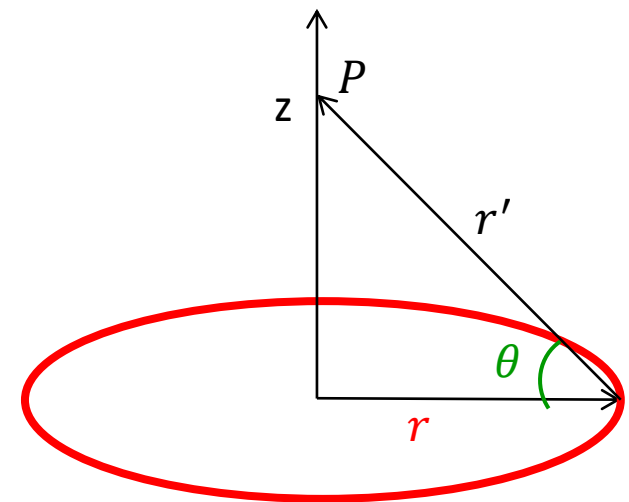
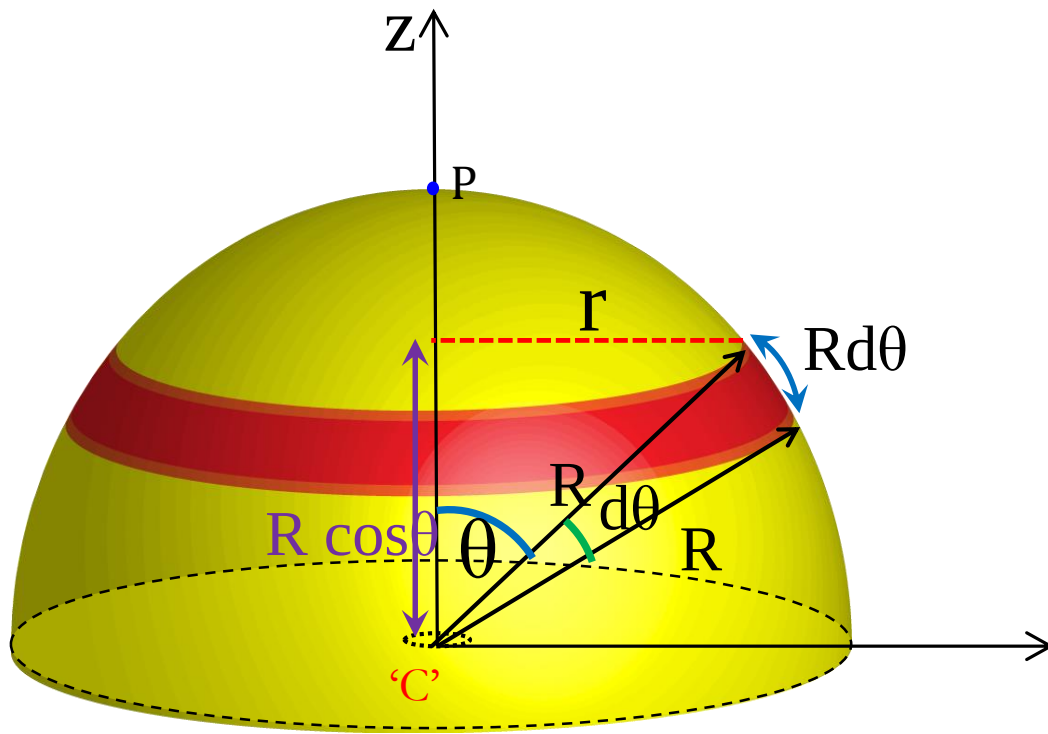


Figure 2

One strip of
hemispherical shell

Sol :

From Biot-Savart law we get magnetic field at 'P' due to current carrying circular loop of radius 'r' is ,

$$\vec{B} = \frac{I \mu_o}{2} \frac{r^2}{(r^2 + z^2)^{3/2}} \hat{z}$$

Hemisphere can be considered as a heap of such many current loops .

From figure 1 ,

$$z = R - R \cos \theta \quad \text{and} \quad r = R \sin \theta$$

Current on the strip $I = (\text{surface charge density}) \times (\text{area}) \times (\text{angular velocity})$

$$\text{or, } I = \sigma (R \sin \theta) (R d\theta) \omega$$

Magnetic field due to this strip is

$$d\vec{B} = \frac{I \mu_o}{2} \frac{(R \sin \theta)^2}{(R^2 \sin^2 \theta + R^2 - 2R^2 \cos \theta + R^2 \cos^2 \theta)^{3/2}} \hat{z}$$

or, $d\vec{B} = \frac{I \mu_o}{2} \frac{R^2 \sin^2 \theta}{(2R^2 - 2R^2 \cos \theta)^{3/2}} \hat{z}$ where $I = \sigma(R \sin \theta)(R d\theta)\omega$

$$d\vec{B} = \frac{\mu_o (\sigma \omega R^2 \sin \theta d\theta)}{2} \frac{R^2 \sin^2 \theta}{2^{3/2} R^3 (1 - \cos \theta)^{3/2}} \hat{z}$$

$$d\vec{B} = \frac{\mu_o \sigma \omega}{2} \frac{R \sin^3 \theta d\theta}{2^{3/2} (1 - \cos \theta)^{3/2}} \hat{z}$$

Magnetic field at point P due to whole hemisphere is ,

$$\vec{B} = \int d\vec{B} = \frac{\mu_0 \sigma \omega}{2^{5/2}} \int_0^{\pi/2} \frac{R \sin^3 \theta d\theta}{(1 - \cos \theta)^{3/2}} \hat{z}$$

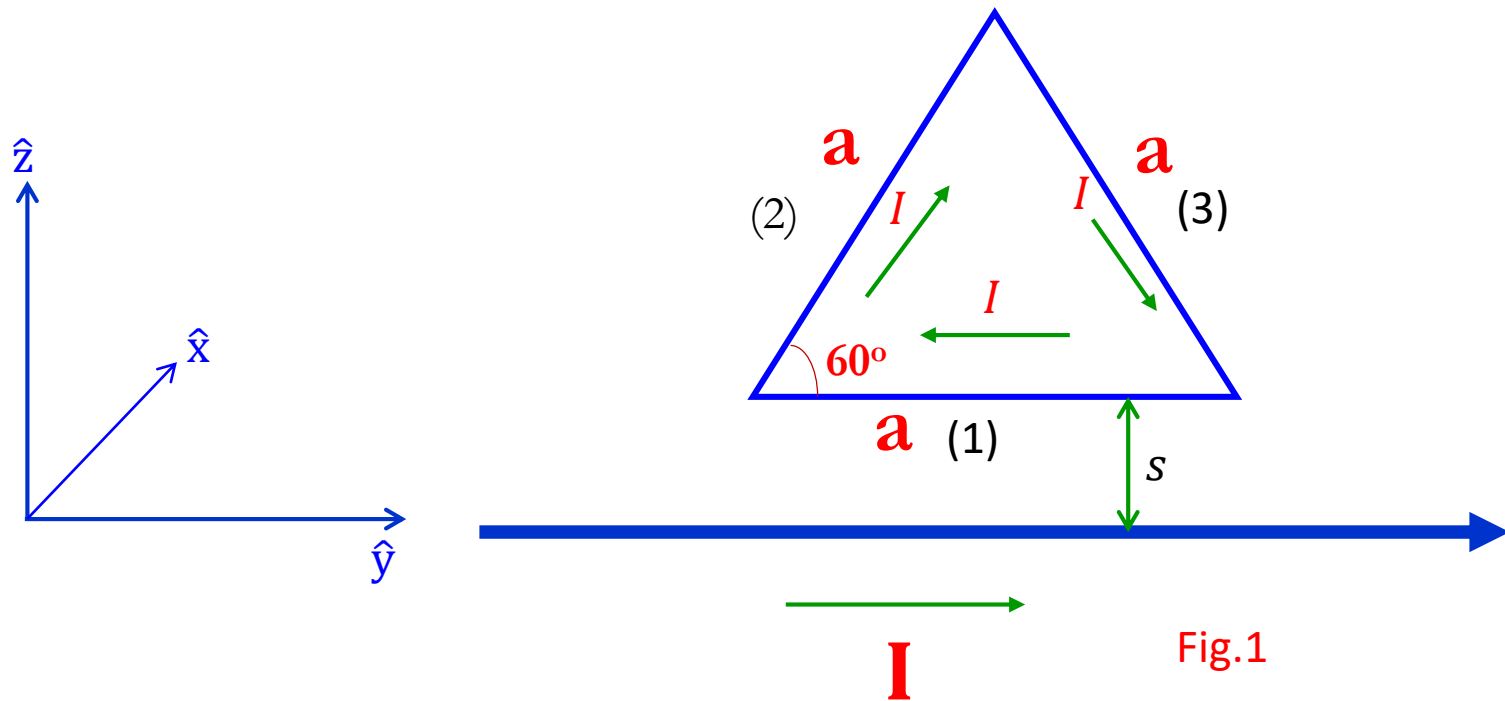
$$\vec{B} = \frac{\mu_0 \sigma \omega R}{2^{5/2}} \int_0^{\pi/2} \frac{(1 - \cos^2 \theta) \sin \theta d\theta}{(1 - \cos \theta)^{3/2}} \hat{z}$$

$$\vec{B} = \frac{\mu_0 \sigma \omega R}{2^{5/2}} \int_0^{\pi/2} \frac{(1 + \cos \theta) \sin \theta d\theta}{(1 - \cos \theta)^{1/2}} \hat{z}$$

$$\vec{B} = \frac{\mu_0 \sigma \omega R}{2^{\frac{5}{2}}} \left(\frac{10}{3} \right) \hat{z}$$

Example

Find the force on the triangular loop shown in figure .Both loops carry a current I .



Let , the straight current carrying wire is placed along y -axis . And triangular loop is placed in the yz -plane . In triangular loop current flows in clockwise.

The force on an element of length $d\vec{l}$ is ,

$$\vec{F} = \int I d\vec{l} \times \vec{B}$$

Force on segment (1) of length a is

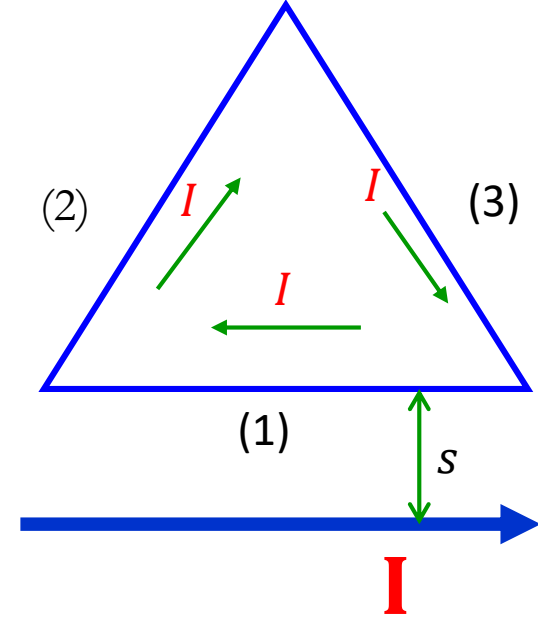
$$\vec{F}_1 = \int I d\vec{l}_1 \times \vec{B}$$

Where $d\vec{l}_1 = -\hat{y}dy$

Magnetic field due to straight current carrying wire at a perpendicular distance s is , $\vec{B} = \frac{\mu_0 I}{2\pi s} \hat{x}$

$$\begin{aligned} \vec{F}_1 &= \int I(-\hat{y}dy \times \frac{\mu_0 I}{2\pi s} \hat{x}) \\ &= \hat{z} \frac{\mu_0 I^2}{2\pi s} \int_0^a dy \end{aligned}$$

$$\vec{F}_1 = \hat{z} \frac{\mu_0 I^2 a}{2\pi s} \quad \text{-----} \quad (1)$$



Consider segment (2)

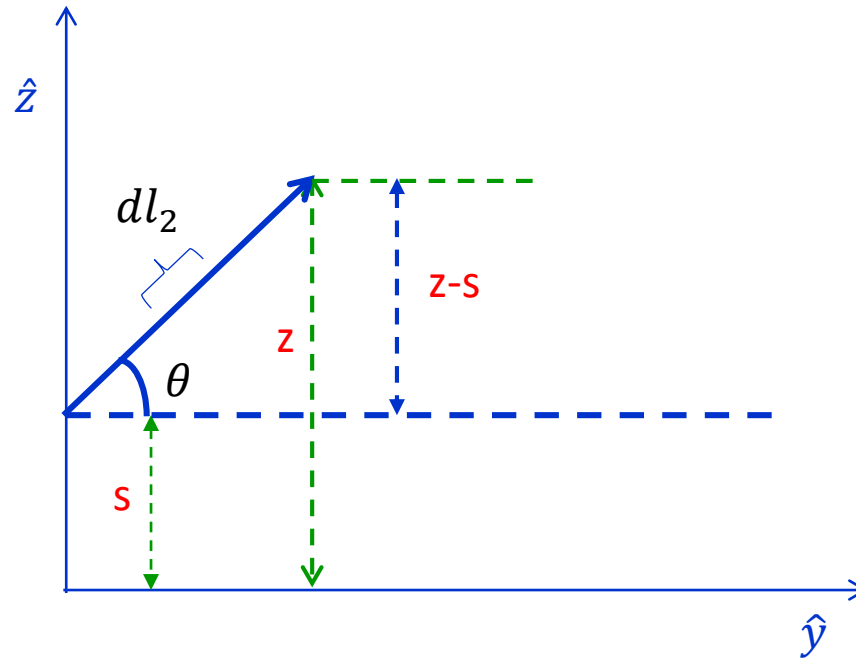


Fig.2

$$\vec{dl}_2 = \hat{y}dy + \hat{z}dz$$

Force on segment (2) of length a is $\vec{F}_2 = \int I d\vec{l}_2 \times \vec{B}$

Magnetic field due to straight current carrying wire at a perpendicular distance z is , $\vec{B} = \frac{\mu_0 I}{2\pi z} \hat{x}$

$$\begin{aligned}\vec{F}_2 &= \int I(\hat{y}dy + \hat{z}dz) \times \frac{\mu_0 I}{2\pi z} \hat{x} \\ &= \frac{\mu_0 I^2}{2\pi} \int -\frac{\hat{z}dy}{z} + \hat{y}(dz/z) \quad \text{---} \quad (ii)\end{aligned}$$

The integral over y is not as simple as it may seem because the value of z depends on y . The section of wire is a straight line and z can be expressed as $\mathbf{z} = \mathbf{m}y + \mathbf{c}$

Initial point of line $\mathbf{y} = \mathbf{0} \quad \mathbf{z} = \mathbf{s} \quad \rightarrow \mathbf{c} = \mathbf{s}$

Final point of line $\mathbf{y} = \frac{\mathbf{a}}{2} \quad \mathbf{z} = \frac{\mathbf{ma}}{2} + \mathbf{s}$

Where $\mathbf{m} = \text{slope of the line} = \frac{dz}{dy} = \tan 60^\circ = \sqrt{3}$ (From fig.2)

So, $\mathbf{z} = \mathbf{y}\sqrt{3} + \mathbf{c} \rightarrow \mathbf{z} = \frac{\sqrt{3}\mathbf{a}}{2} + \mathbf{s} \quad (\text{at } \mathbf{y} = \frac{\mathbf{a}}{2}) \quad \text{--} \quad (iii)$

Substituting the integration limit into equation (ii)

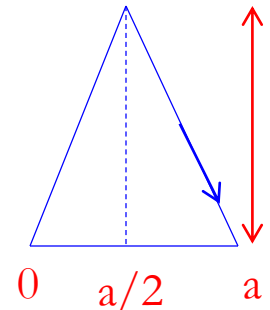
$$\vec{F}_2 = \frac{\mu_0 I^2}{2\pi} \left[-\hat{z} \int_0^{\frac{a}{2}} \frac{dy}{s + \sqrt{3}y} + \hat{y} \int_0^{\frac{\sqrt{3}a}{2} + s} \frac{dz}{z} \right]$$

$$\vec{F}_2 = \frac{\mu_0 I^2}{2\pi} \left[-\hat{z} \frac{1}{\sqrt{3}} \ln\left(1 + \frac{\sqrt{3}a}{2s}\right) + \hat{y} \ln\left(1 + \frac{\sqrt{3}a}{2s}\right) \right] \quad (iv)$$

For segment (3) : in this segment slope is negative , $m = -\sqrt{3} = \tan 60$

Initial point of line $y = \frac{a}{2}$, $z = -\frac{\sqrt{3}a}{2} + c$

But from equation (iii) we know $\rightarrow z = \frac{\sqrt{3}a}{2} + s$



$$\text{or, } -\frac{\sqrt{3}a}{2} + c = \frac{\sqrt{3}a}{2} + s$$

$$\text{or, } c = \sqrt{3}a + s$$

$$d\vec{l}_3 = \hat{y}dy - \hat{z}dz$$

So, $z = my + c$

or, $z = -\sqrt{3}y + \sqrt{3}a + s$

Force on segment (3) of length a is $\vec{F}_3 = \int I d\vec{l}_3 \times \vec{B}$

$$d\vec{l}_3 = \hat{y}dy - \hat{z}dz$$

$$\vec{F}_3 = \int I(\hat{y}dy - \hat{z}dz) \times \frac{\mu_0 I}{2\pi z} \hat{x}$$

$$= \frac{\mu_0 I^2}{2\pi} \int -\frac{\hat{z}dy}{z} - \hat{y}(dz/z)$$

at
 $y = \frac{a}{2}$
 $z = \frac{\sqrt{3}a}{2} + s$

$$\vec{F}_3 = \frac{\mu_0 I^2}{2\pi} \left[-\hat{z} \int_{\frac{a}{2}}^a \frac{dy}{-\sqrt{3}y + \sqrt{3}a + s} - \hat{y} \int_{\frac{\sqrt{3}a}{2} + s}^s \frac{dz}{z} \right]$$

$$\vec{F}_3 = \frac{\mu_0 I^2}{2\pi} \left[-\hat{z} \frac{1}{\sqrt{3}} \ln\left(1 + \frac{\sqrt{3}a}{2s}\right) - \hat{y} \ln\left(1 + \frac{\sqrt{3}a}{2s}\right) \right] \quad (v)$$

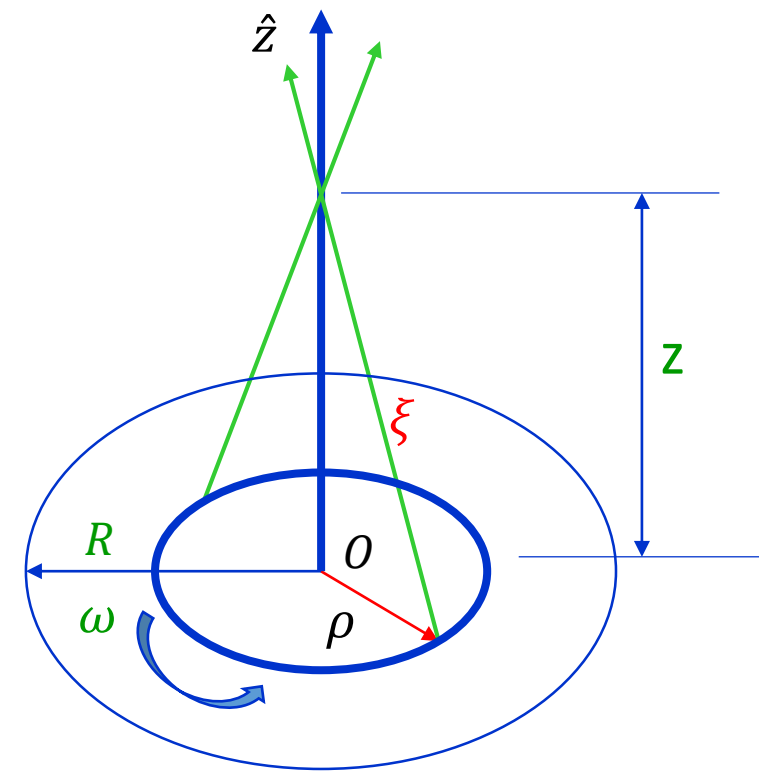
Total force acting on the triangular loop is ,

$$\vec{F} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3$$

$$\vec{F} = \frac{\mu_0 I^2 a}{2\pi s} \hat{z} - \frac{2}{\sqrt{3}} \frac{\mu_0 I^2}{2\pi} \ln\left(1 + \frac{\sqrt{3}a}{2s}\right) \hat{z}$$

Example

A uniformly charged thin disk of charged density σ radius R and thickness $t \ll R$ rotates with an angular velocity ω about the z -axis of symmetry as shown in figure. Find out the magnetic field B at distance z in the vertical axis.



The current density in the disk is given by

$$\mathbf{J} = (\rho_{\text{charge density}}) \vec{v} = \sigma \omega \rho \hat{\phi}$$

Where ρ = is the distance

from the axis of the disk

The ring shaped portion of the disk of radial thickness $d\rho$ thus constitute a current ring $dI = \sigma \omega \rho d\rho$

The $\vec{B}$ field on the z axis due to this current ring is $d\vec{B} = \frac{\mu_0}{2} \frac{dI \rho^2}{\xi^3} \hat{z}$

Note that $\vec{B}$ field on the axis of the circular loop is ,

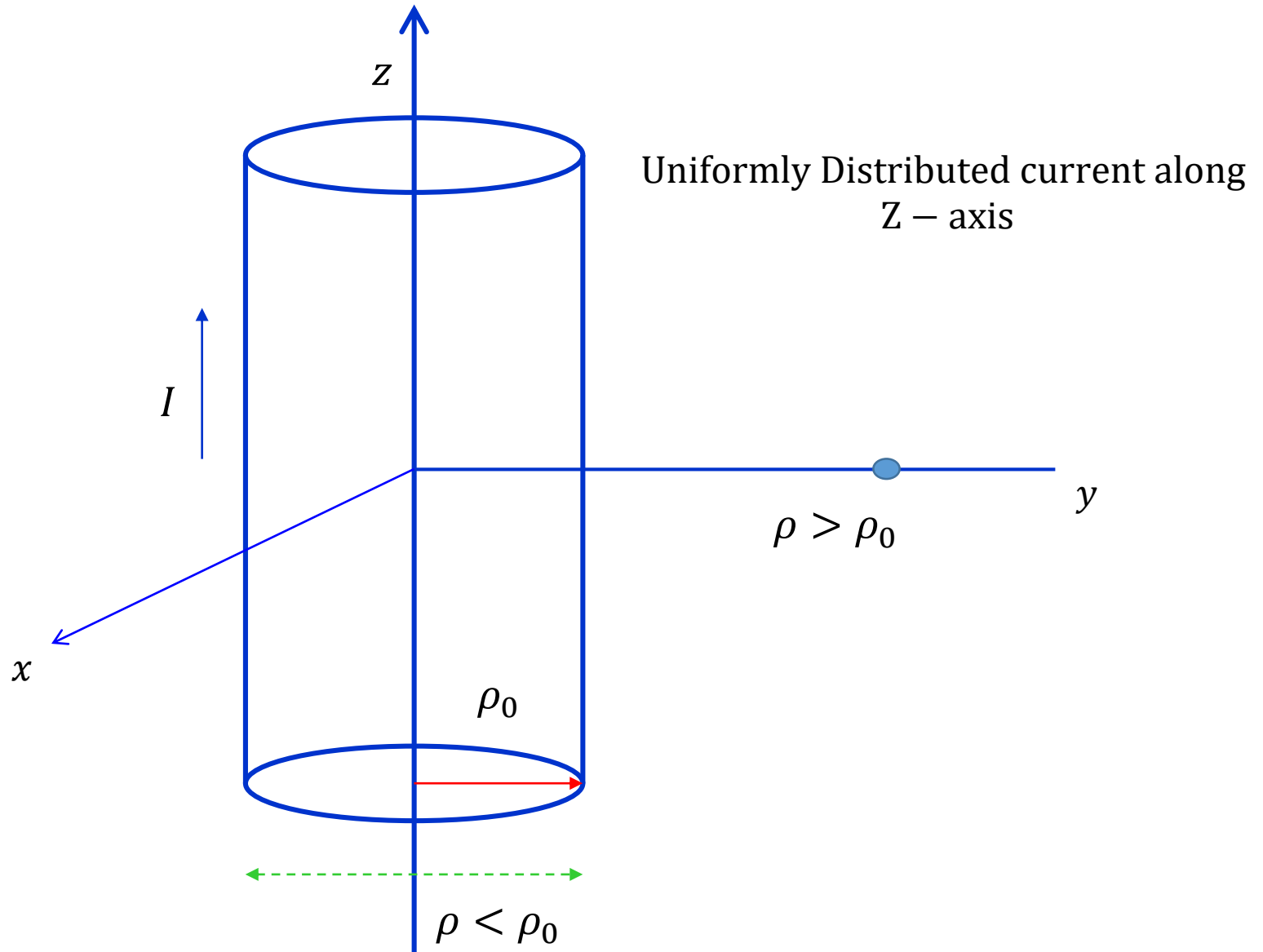
$$\vec{B} = B_z \hat{z} = \frac{\mu_0}{2} \frac{IR^2}{\xi^3} \hat{z} \quad R = \text{circular loop radius}$$

$$\Rightarrow d\vec{B} = \frac{\mu_0}{2} \frac{\sigma \omega \rho^3 d\rho}{\xi^3} \hat{z} \quad \xi^2 = \rho^2 + z^2$$

$$\vec{B} = \int_0^R \left(\frac{\mu_0 \sigma \omega}{2} \right) \frac{\rho^3 d\rho}{(\rho^2 + z^2)^{3/2}} \hat{z}$$

$$\vec{B} = \hat{z} \frac{\mu_0 \sigma \omega}{2} \left[\frac{R^2 + 2z^2}{\sqrt{R^2 + z^2}} - 2|z| \right]$$

Magnetic vector potential due to nonfilamentary current



Consider long cylindrical conductor of radius ρ_0 , constant current flowing in I

$$\nabla^2 A_z = -\mu_0 J \text{ where } J = \frac{I}{\pi \rho_0^2}$$

Note : no current sources in the x and y directions . A_x and A_y may be taken to be zero .

$A_z \rightarrow$ depends only on the distance from the axis of the conductor .

A_z satisfies the equation ,

$$\frac{1}{\rho} \frac{d}{d\rho} \left(\rho \frac{dA_z}{d\rho} \right) = -\mu_0 J$$

$$\frac{d}{d\rho} \left(\rho \frac{dA_z}{d\rho} \right) = -\mu_0 J \rho$$

Integrating , $\rho \frac{dA_z}{d\rho} = - \int \mu_0 \rho J d\rho$

$$= - \frac{\mu_0 \rho^2 J}{2} + C_1$$

$$dA_z = - \frac{\mu_0 \rho J}{2} d\rho + \frac{C_1}{\rho} d\rho$$

Integrating , $A_z = - \frac{\mu_0 J}{2} \int \rho d\rho + \int \frac{C_1}{\rho} d\rho + C$

$$A_z = - \frac{\mu_0 J \rho^2}{4} + \ln(\rho) C_1 + C \quad \text{where } J = \frac{I}{\pi \rho_0^2}$$

$$A_z = - \frac{\mu_0 \rho^2}{4} \frac{I}{\pi \rho_0^2} + \ln(\rho) C_1 + C \text{ ---- (a) for } \rho < \rho_0$$

$$A_z = \ln(\rho) D_1 + D \quad \text{for } \rho > \rho_0 \text{ (out side } I = 0) \text{ ---- } (b)$$

Here we use boundary conditions on potentials (Magnetic Scalar Potential) to evaluate these constants

The potential inside the conductor must be finite because the current is uniformly distributed (no filamentary current at the axis) then C_1 must be taken to zero ((at the centre $\ln [\rho=0]$ which is undefined). ' $\mathbf{A}$ ' should be continuous at $\rho = \rho_0$

$$A_z = -\frac{\mu_0 \rho^2}{4} \frac{I}{\pi \rho_0^2} + C \quad \leftarrow \text{from Eq (a) for } C_1 = 0 \text{ for } \rho < \rho_0$$

For $\rho > \rho_0$

$$A_{1t} = A_{2t}$$

From equations (a) & (b)

$$-\frac{\mu_0 I}{4\pi} + C = \ln(\rho_0) D_1 + D \quad (1)$$

$$B = \nabla \times A_z = \nabla \times (\ln(\rho) D_1 + D)$$

$$B = -(D_1 / \rho) \hat{\phi}$$

Now apply Ampere's law to a circle of radius $\rho > \rho_0$

$$\oint \vec{B} \cdot d\vec{l} = - \int_0^{2\pi} (D_1 / \rho) \rho d\phi = \mu_0 I$$

$$\Rightarrow -D_1 2\pi = \mu_0 I$$

$$\Rightarrow D_1 = -\frac{\mu_0 I}{2\pi}$$

Substituting D_1 in equation (1)

$$D = -\frac{\mu_0 I}{4\pi} + C + \frac{\mu_0 I}{2\pi} \ln(\rho_0)$$

Substituting D_1 and D both in equation (b)



$$A_z = \ln(\rho) D_1 + D$$

$$A_z = -\frac{\mu_0 I}{2\pi} \ln(\rho) + \frac{\mu_0 I}{2\pi} \ln(\rho_0) - \frac{\mu_0 I}{4\pi} + C \quad \text{for } \rho > \rho_0$$

$$A_z = -\frac{\mu_0 I}{2\pi} (-\ln(\rho_0) + \ln(\rho)) - \frac{\mu_0 I}{4\pi} + C \quad \text{for } \rho > \rho_0$$

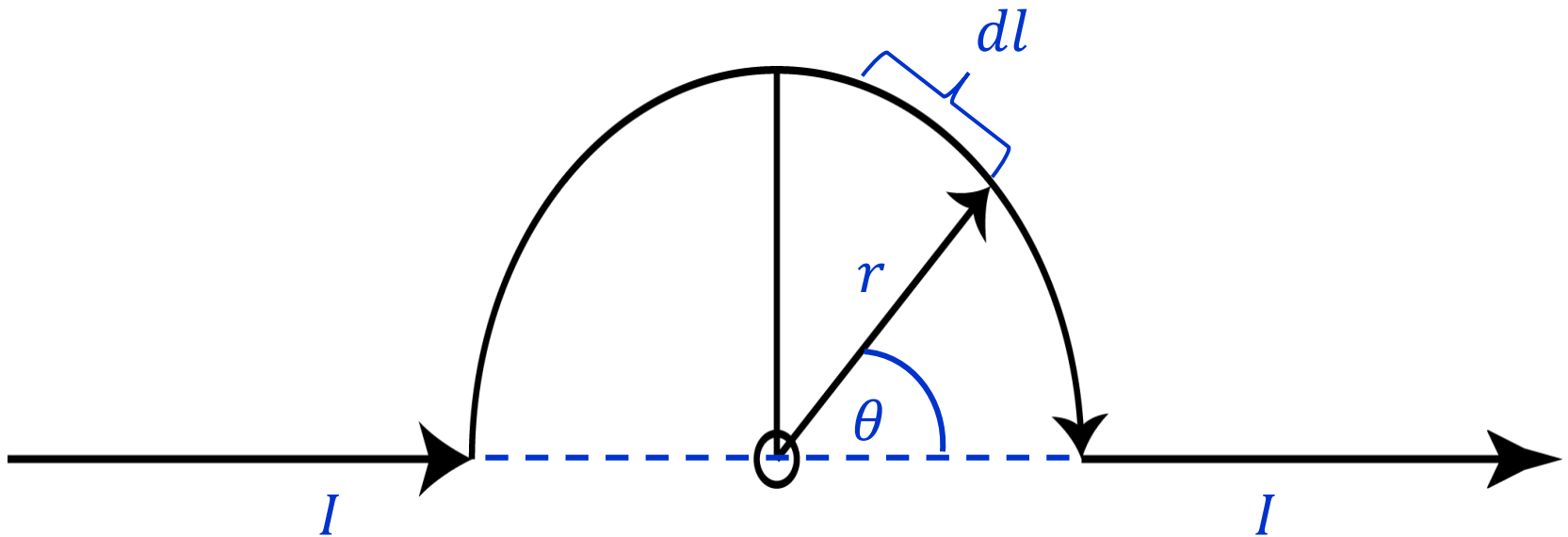
$$A_z = -\frac{\mu_0 I}{4\pi} (2 \ln \frac{\rho}{\rho_0} + 1) + C \quad \text{for } \rho > \rho_0$$

$$A_z = -\frac{\mu_0 I}{4\pi} \left(\frac{\rho}{\rho_0}\right)^2 + C \quad B = \frac{\mu_0 I}{2\pi} \frac{\rho}{\rho_0^2} \hat{\phi} \quad \text{for } \rho < \rho_0$$

$$A_z = -\frac{\mu_0 I}{4\pi} (2 \ln \frac{\rho}{\rho_0} + 1) + C \quad B = \frac{\mu_0 I}{2\pi \rho} \hat{\phi} \quad \text{for } \rho > \rho_0$$

Example

An infinitely long wire carries a current $I = 1\text{ A}$. It is bent so as to have a semi-circular detour around the origin, with radius 1 cm . Calculate the magnetic field at the origin.



The straight parts of the wire do not contribute to the magnetic field at O .

Since for then $I d\vec{l} \times \vec{r} = 0$

we need only to consider the contribution of a semi-circular part . The magnetic field at O produced by a current element $I d\vec{l}$ is

$$dB = \frac{\mu_0}{4\pi} \frac{I d\vec{l} \times \vec{r}}{r^3} = \frac{\mu_0}{4\pi} \frac{|dl||r|\sin 90^\circ I}{r^3}$$

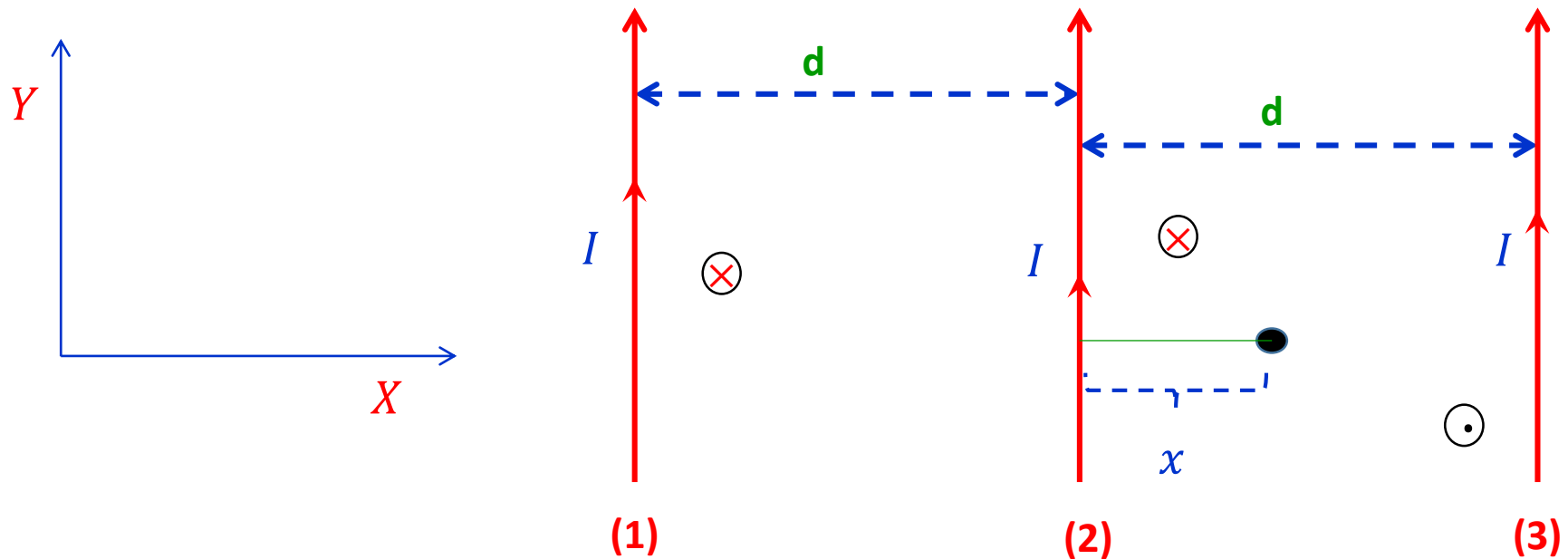
As $I d\vec{l}$ and $\vec{r}$ are mutually perpendicular for the semi-circular wire , $d\vec{B}$ is always pointing into the page .

$$B = \int dB = \frac{\mu_0 I}{4\pi r^2} \int_0^\pi r d\theta = \frac{\mu_0 I}{4r} = B = 3.14 \times 10^{-5} T$$

Example

Consider three straight, infinitely long, equally spaced wires (with zero radius), each carrying a current I in the same direction.

- (a) Calculate the location of two zero in the magnetic field.
- (b) Sketch the magnetic field line pattern.
- (c) If the middle wire is rigidly displaced a very small distance x ($x \ll d$) upward while the other two wires held fixed, describe qualitatively the subsequent motion of the wire.



(a) Assume the three wires to be coplanar , then the points of zero magnetic field must also be located in the same plane . Let the distance of such a point from the middle wire (2) be x . Then the distance of this point from the wire (1) and wire (3) are $(d+x)$ and $(d-x)$ respectively . Applying ampere's law we obtain,

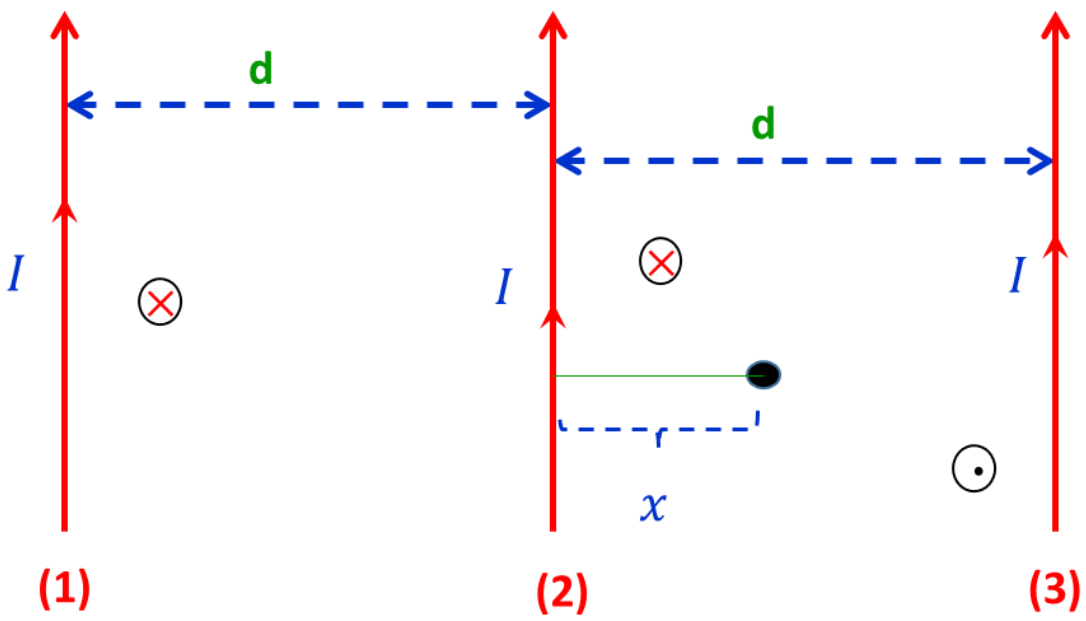
$$\int \vec{B} \cdot d\vec{l} = \mu_0 I \Rightarrow B = \frac{\mu_0 I}{2\pi s}$$

For a point of zero magnetic field,

$$\frac{\mu_0 I}{2\pi(d - x)} = \frac{\mu_0 I}{2\pi x} + \frac{\mu_0 I}{2\pi(d + x)}$$

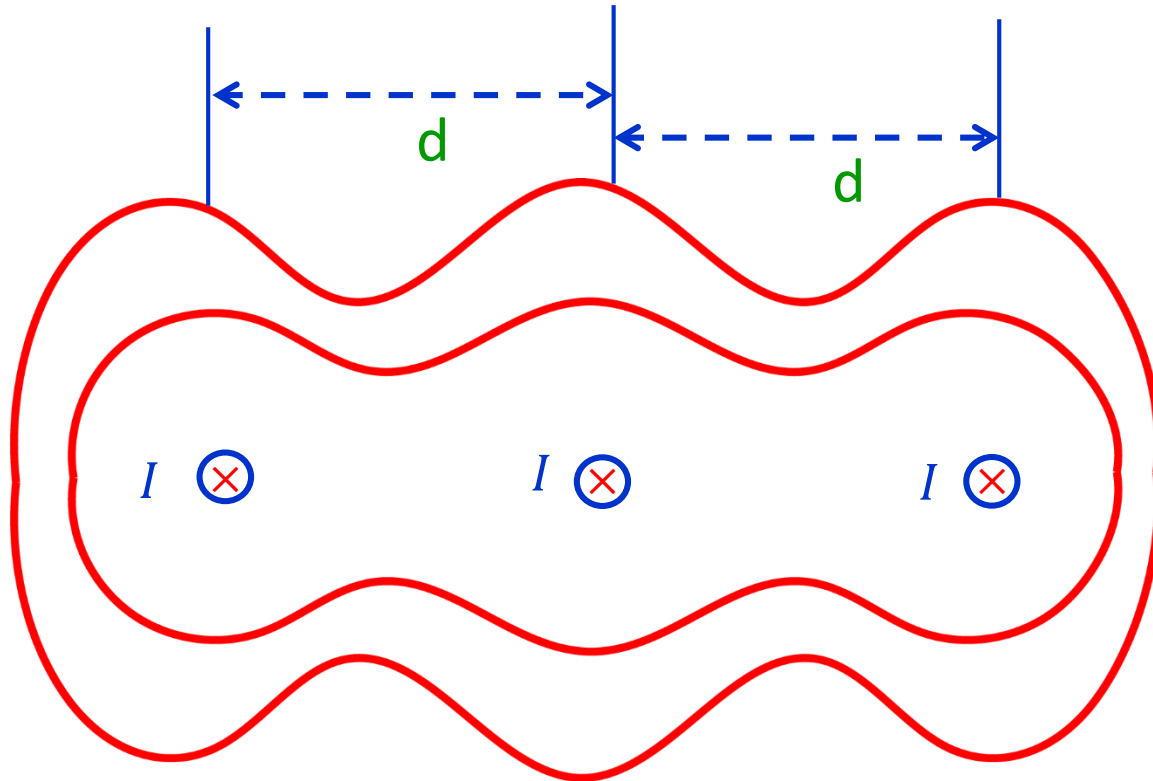
↓

$$x = \pm \frac{1}{\sqrt{3}} d$$



Correspondingly to two points located between the middle wire and each of the other two wires , both having distance $\frac{1}{\sqrt{3}} d$ from the middle wire .

(b) The magnetic field lines are



(c) When the middle wire is displaced a small distance x in the same plane , the force per unit length on the wire is , $\vec{F} = Id\vec{l} \times \vec{B}$

$$\Rightarrow f = \frac{F}{L} = IB$$

the resultant force per unit length on the wire is

$$f = \frac{\mu_0 I^2}{2\pi(d+x)} - \frac{\mu_0 I^2}{2\pi(d-x)}$$

As $x \ll d$, we obtain approximately , $f \approx -\frac{\mu_0 I^2}{\pi d^2} x$

This equation looks like simple harmonic oscillator $F = -\omega^2 x$

$$\therefore \omega = \sqrt{\frac{\mu_0 I^2}{\pi d^2}}$$

$\Rightarrow$ the force is proportional but opposite to the displacement . Hence , the motion is simple harmonic about the equilibrium position with a period

$$T = 2\pi \sqrt{\frac{\pi m d}{\mu_0 I}} \quad \text{where } m \text{ is the mass per unit length of the wire .}$$

Inhomogeneous magnetic field

Stern-Gerlach Experiment

Beam of particles through an inhomogeneous magnetic field and observing their deflection leads to [angular momentum](#) Quantization.

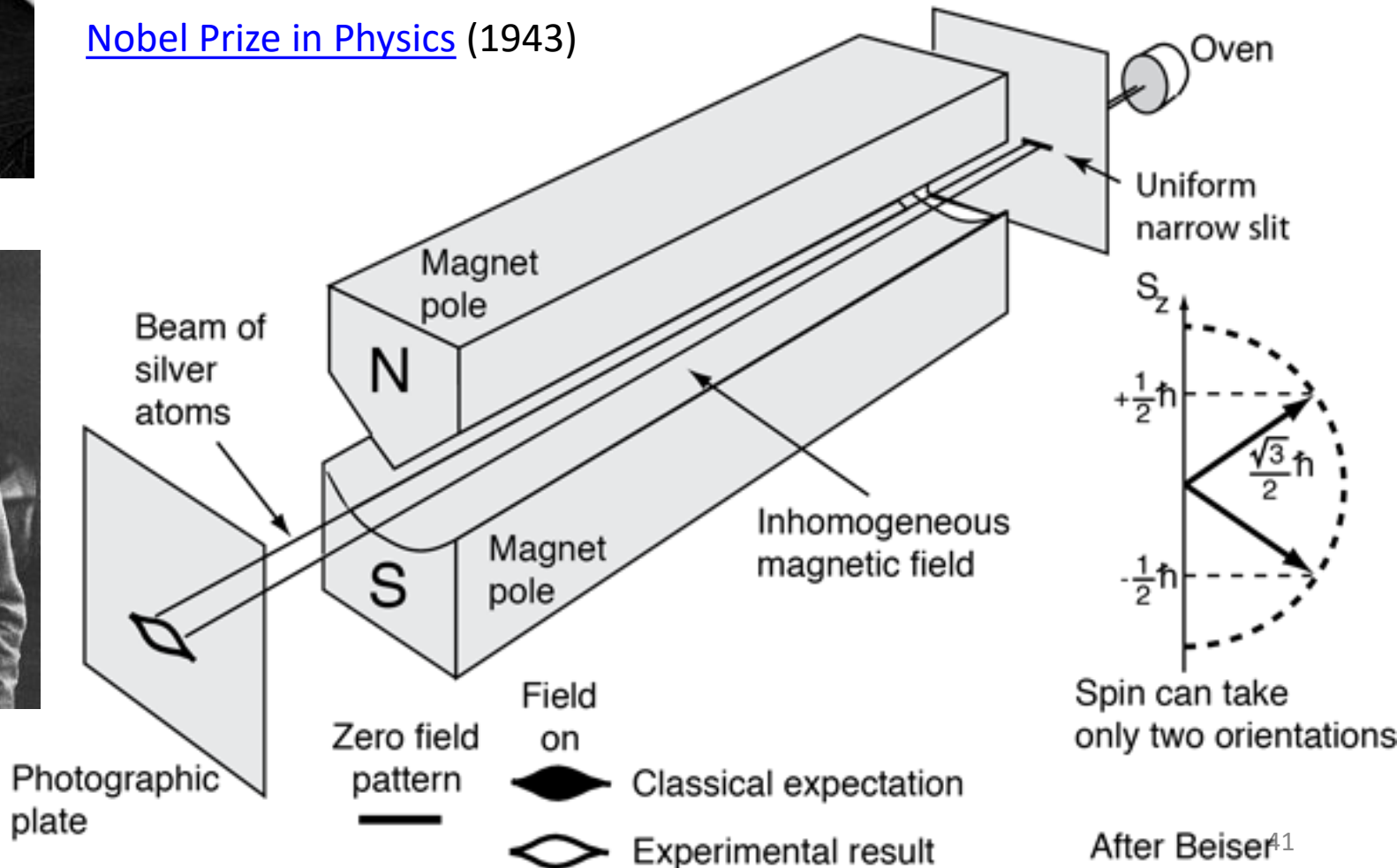
[Nobel Prize in Physics](#) (1943)



Otto Stern



Walter Gerlach



Otto Stern



Walter Gerlach



The Stern–Gerlach experiment involves sending a beam of particles through an inhomogeneous magnetic field and observing their deflection. The results show that particles possess an intrinsic [angular momentum](#) that is most closely analogous to the angular momentum of a classically spinning object, but that takes only certain quantized values

[Nobel Prize in Physics](#) (1943)

The experiment is normally conducted using electrically neutral particles or atoms. This avoids the large deflection to the orbit of a charged particle moving through a magnetic field and allows spin-dependent effects to dominate. If the particle is treated as a classical spinning [dipole](#), it will [precess](#) in a [magnetic field](#) because of the torque that the magnetic field exerts on the dipole (see [torque-induced precession](#)). If it moves through a homogeneous magnetic field, the forces exerted on opposite ends of the dipole cancel each other out and the trajectory of the particle is unaffected.

However, if the magnetic field is inhomogeneous then the force on one end of the dipole will be slightly greater than the opposing force on the other end, so that there is a net force which deflects the particle's trajectory. If the particles were classical spinning objects, one would expect the distribution of their spin angular momentum vectors to be [random and continuous](#).

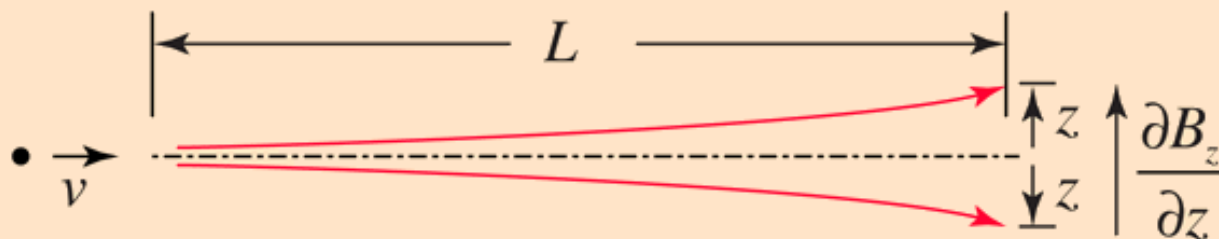
$$U = -\mu \cdot B = -\mu_B \frac{g}{2} B_z = \pm \mu_B B_z$$

where g is the [electron spin g-factor](#) and μ_B is the [Bohr magneton](#).

Using the relationship of [force to potential energy](#) gives

$$F_z = -\frac{\partial U}{\partial z} = \pm \mu_B \frac{\partial B_z}{\partial z}$$

The deflection can be shown to be proportional to the spin and to the magnitude of the magnetic field gradient. It is inversely proportional to the particle kinetic energy.



$$z = \frac{1}{2} a t^2 = \frac{1}{2} \frac{F}{m} \left[\frac{L}{v} \right]^2 = \pm \frac{\mu_B L^2}{4KE} \frac{\partial B_z}{\partial z}$$

Electron Spin Magnetic Moment

Since the electron displays an [intrinsic angular momentum](#), one might expect a [magnetic moment](#) which follows the form of that for an [electron orbit](#). The z-component of magnetic moment associated with the [electron spin](#) would then be expected to be

$$\mu_z = \pm \frac{1}{2} \mu_B$$

but the measured value turns out to be about twice that. The measured value is written

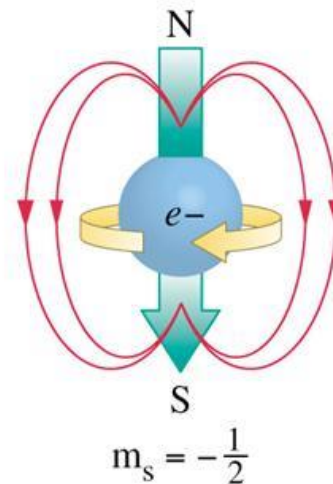
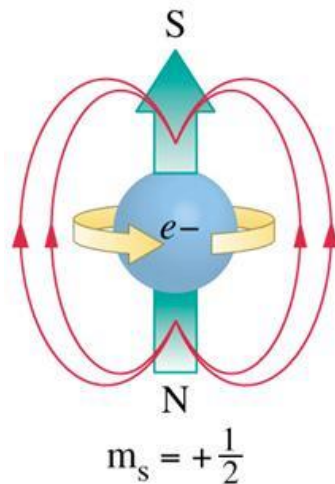
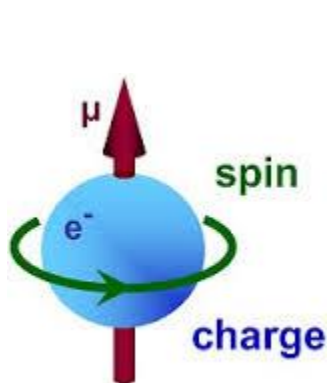
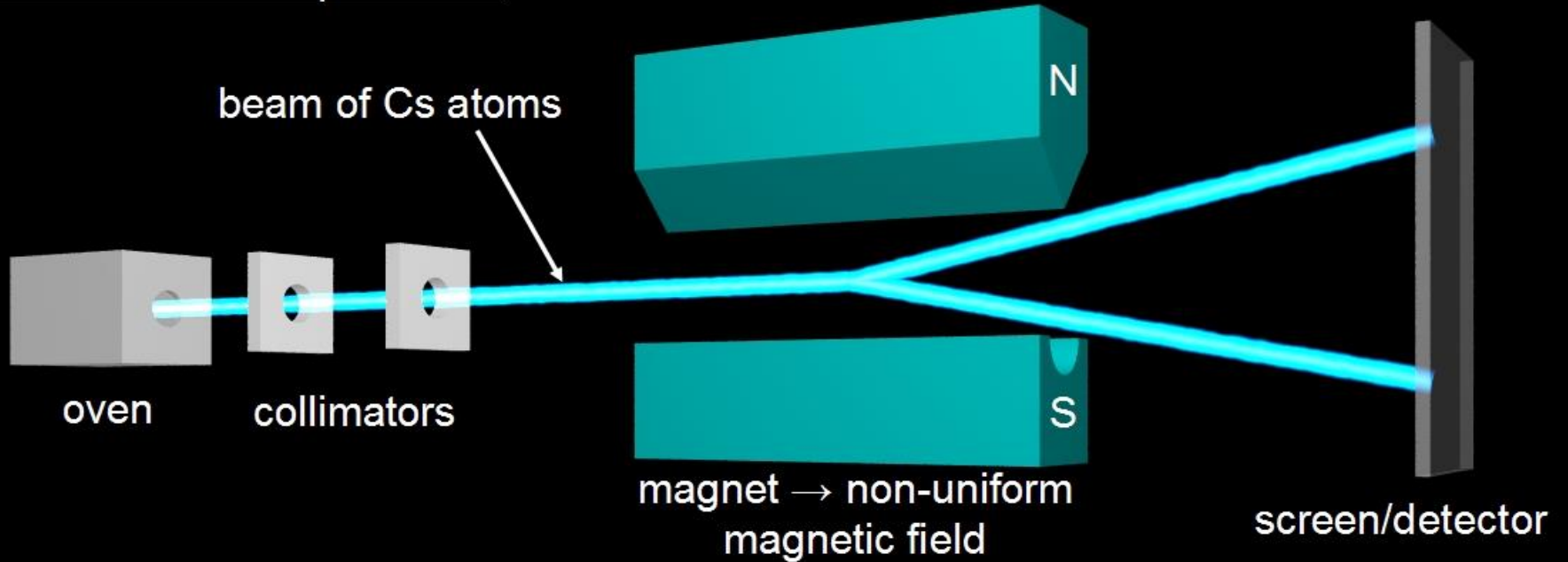
$$\mu_z = \pm \frac{1}{2} g \mu_B$$

where g is called the gyromagnetic ratio and the [electron spin g-factor](#) has the value $g = 2.00232$ and $g=1$ for orbital angular momentum. The precise value of g was predicted by relativistic quantum mechanics in the Dirac equation and was measured in the [Lamb shift](#) experiment. A natural constant which arises in the treatment of magnetic effects is called the [Bohr magneton](#). The magnetic moment is usually expressed as a multiple of the Bohr magneton.

$$\mu_B = \frac{e\hbar}{2m_e} = 9.2740154 \times 10^{-24} \text{ J / T} = 5.7883826 \times 10^{-5} \text{ eV / T}$$

Bohr magneton

Stern-Gerlach experiment



Angular momentum

In 1927, T.E. Phipps and J.B. Taylor reproduced the effect using [hydrogen](#) atoms in their [ground state](#), thereby eliminating any doubts that may have been caused by the use of [silver](#) atoms.

FEBRUARY, 1927

PHYSICAL REVIEW

VOLUME 29

THE MAGNETIC MOMENT OF THE HYDROGEN ATOM

BY T. E. PHIPPS AND J. B. TAYLOR

ABSTRACT

The magnetic moment of the hydrogen atom has been investigated by the atomic ray method introduced by Stern and Gerlach. Atomic hydrogen formed in a discharge tube by the method of R. W. Wood was first used. The ray was formed in a special all-glass slit system of three slits sealed to the discharge tube. The ray was detected by the reduction resulting on contact with a target coated with molybdenum trioxide. A sharply defined blue line against a white background was the result. In the magnetic field the ray was separated into two branching rays. There was also evidence of a central undeviated ray which is believed to be due to hydrogen active chemically but probably not in the atomic state. From a measurement of the deflection the magnetic moment of the hydrogen atom was calculated to be one Bohr magneton within the limits of experimental error. This result is of interest because of the questions raised by the new quantum mechanics of Heisenberg, Born, and Jordan, and by the spinning electron theory of Uhlenbeck and Goudsmit. Atomic hydrogen formed by the hot filament method of Langmuir was next used. The increased velocity of the atoms in this case resulted in less separation of the ray, but a deflection was distinctly recorded. Finally, the product formed on exposing a mixture of mercury vapor and hydrogen to ultra-violet light was investigated. This is believed by Cario and Franck, Taylor, and others to be atomic hydrogen. Thus far attempts to form a ray which will reduce the target have been unsuccessful. The reasons for this are being investigated.

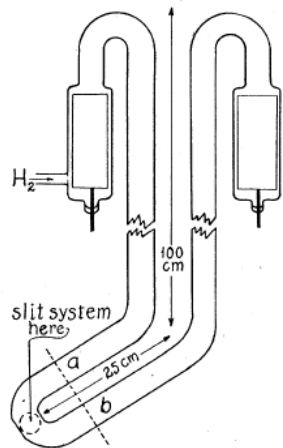


Fig. 1.

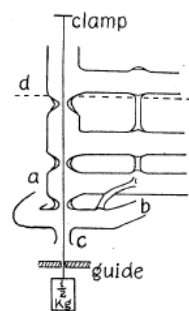


Fig. 2.

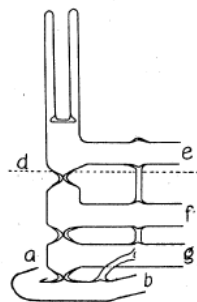


Fig. 1. The discharge tube.

Fig. 2. Method of preparing the glass slit syst

Fig. 3. Completed slit system, showing preliminary

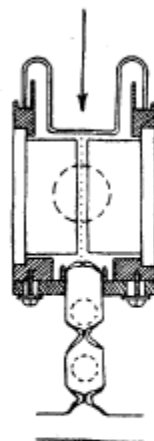


Fig. 4.

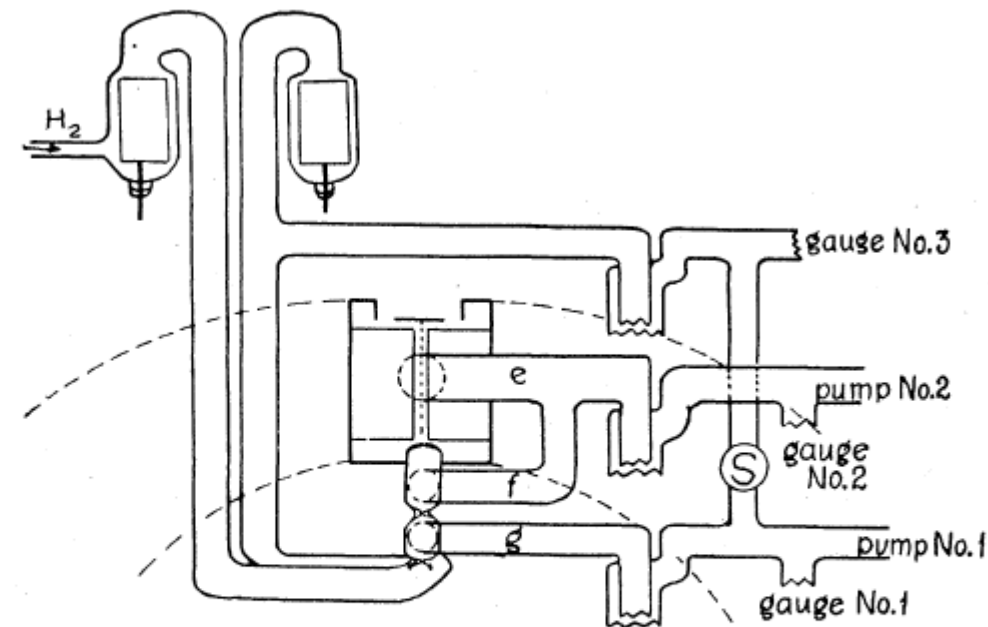
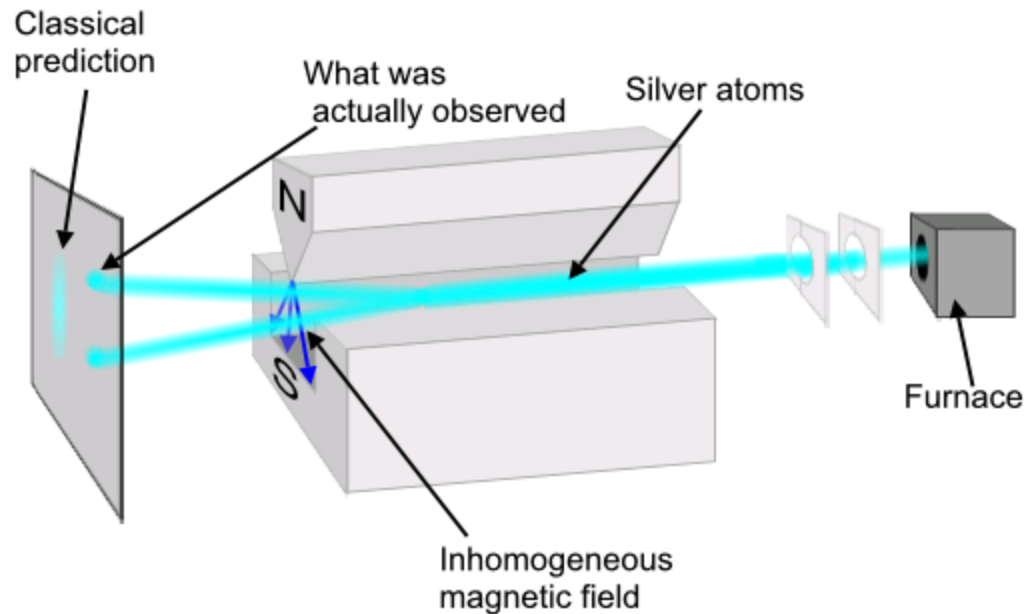


Fig. 5.

Fig. 4. Pole-piece box with slits and target attached.

Fig. 5. Diagrammatic sketch of complete connections. Dotted lines indicate position of electromagnet.



Since the electron was [quantized](#) to be only in certain positions in space, the separation into distinct orbits was referred to as [space quantization](#). The Stern–Gerlach experiment was meant to test the [Bohr–Sommerfeld hypothesis](#) that the direction of the angular momentum of a silver atom is quantized

Note that the experiment was performed several years before [Uhlenbeck](#) and [Goudsmit](#) formulated their hypothesis of the existence of the [electron spin](#). Even though the result of the Stern–Gerlach experiment has later turned out to be in agreement with the predictions of quantum mechanics for a $\text{spin-}\frac{1}{2}$ particle, the experiment should be seen as a corroboration of the [Bohr–Sommerfeld theory](#)